Airbus Challenge #5 KATL · KDFW 3–12 h ahead development run, 31 Oct 2026

Hub Capacity Watch

For a hub flow coordinator, 3 to 12 hours before a bank: will arrival throughput drop, and which departures risk a delay of more than 45 minutes? Built end to end on real BTS flights and IEM METAR/TAF weather. The 2025 test is still sealed.

TARGET What we forecast

- **Arrival throughput (AAR proxy):** BTS wheels-on per UTC hour. No true AAR exists, and this count reflects demand as well as capacity.
- **Severe departure delay:** probability that delay is over 45 min, *given that the flight operates*. Cancellation is reported separately.
- Every forecast has an issue time T. Features use only METARs observed and TAFs issued by T, with wind resolved into headwind, tailwind and crosswind for each runway.

PIPELINE Run on real data

`download → clean → requests → iem → targets → features + demand + slices → baselines + flight_baselines → train → calibrate → evaluate → failures → recommend → explain → dashboard`

STAGE	OUTPUT	CHECK
clean	3,351,383 flights	0.99999 match with the provided CSV; about 400 rows quarantined
iem	52,967 METARs · 20,858 TAFs	no METAR gap over 3 h in any month
requests	6,757,796 forecasts	4 horizons × (flights + airport-hours), totals reconcile
targets	45,360 airport-hours	60 excluded under P12, each kept as null with a reason
slices	thunderstorm 1.8% / 2.0%	IFR 9.0% (KATL) / 5.6% (KDFW) of hours, for evaluation only

RESULTS

Severe delay: signal, but not alert-grade

H2 2024, NOT USED FOR FITTING	KATL	KDFW
Share of flights delayed over 45 min	8.9%	12.0%
PR-AUC (ranking quality)	0.234 (2.6× chance)	0.214 (1.8× chance)
Brier score, model vs base rate	0.076 vs ≈0.081	0.102 vs ≈0.106
"Act" threshold (50% precision)	not reachable	not reachable
"Watch" threshold (30% precision)	not reachable	p ≥ 0.11 · recall 6.8% · ≈74 flags/day

Thresholds were chosen on Oct–Dec 2024 after calibrating on Jul–Sep 2024 (policy WP-D7). Top drivers after month and hour: TAF thunderstorm, runway tailwind, ceiling. These are associations, not causes.

RESULTS

Arrival throughput: 18–26% lower error than the best baseline

H2 2024 RMSE, TOUCHDOWNS/HOUR [95% CI]	KATL	KDFW	POOLED
LightGBM	**6.92** [6.11, 7.88]	**9.17** [8.57, 9.82]	**8.12** [7.61, 8.6
Schedule-only	8.46 [7.79, 9.23]	13.00 [12.56, 13.43]	10.96 [10.61, 1
Climatology	8.78 [8.17, 9.49]	13.10 [12.62, 13.61]	11.14 [10.77, 1
Seasonal persistence	10.63 [9.73, 11.65]	11.51 [10.48, 12.56]	11.08 [10.28, 1

Persistence (simulated feed)	39.12	37.69	38.41

- Scored on 35,176 airport-hour forecasts (all horizons) from H2 2024, which no model was fitted or tuned on. On H1 2024 the pattern holds (pooled 6.77 vs 8.46 for climatology), but early stopping used H1, so it is partly in-sample.
- **Error barely changes from 3 h to 12 h** (KDFW 9.12 to 9.21). The model learns mostly from the schedule, which is known at every horizon. Weather features (TAF ceiling, freezing precipitation) rank lower in importance.
- At KATL the confidence intervals of LightGBM and schedule-only touch, so that gain is less certain than at KDFW.

FAILURE MODES

Where it fails, and how the user knows

> ### The product abstains instead of crying wolf.
>
> No flight-level "act" alert is issued, because no threshold met the precision target. The coordinator sees the reason `precision_target_not_reached`, not a silent all-clear.

- **Throughput "act" alerts mostly flag quiet hours.** Against the clear-weather ceiling, they fire on about 56% of hours in Oct–Dec 2024. Across the 2024 replay, "act" covers 53% of KATL forecasts and 70% of KDFW forecasts. Next step: compare against scheduled arrivals in busy hours.
- **Bad weather is rare.** Thunderstorms occur in about 2% of hours, so storm-hour metrics carry wide intervals.
- **Missing or stale inputs are visible.** Every forecast carries a `status` and `reason_codes` (for example `metar_stale`, `taf_uncovered`, `source_hour_partial`). Unavailable forecasts are never filled silently.
- **Simulated assumptions:** schedules are known in hindsight, the persistence feed has zero latency, and labels are treated as known at the fit cutoff.

TRUST

Guards against fooling ourselves

GUARD	STATE
Data after T cannot change a feature at T (tested)	in place
Each model is scored only from its own fit cutoff, on the same rows as its baselines	in place

2025 test: one claim per experiment, in a shared ledger plus a committed mirror	`sealed, not run`
LLM brief: restates code-computed numbers only, checked against the facts, with a template fallback	`in place`
Independent leakage audit (gate G3)	`paused by lead`
Throughput early stopping uses H1 2024, so validation is partly in-sample	`disclosed`

LIVE DEMO Drive the replay

```
uv run python -m airbus_challenge.recommend --run lgbm_throughput_dev
uv run streamlit run app/dashboard.py
```

- Pick KDFW and slide the issue time to a thunderstorm afternoon. Show the forecast with its p10–p90 interval, the METAR and TAF ages at T, and the advisory level with its WP-D7 thresholds.
- Pick a quiet overnight hour and show the throughput "act" firing on low demand. This is the known failure mode.
- Find an hour with a stale METAR and show the advisory abstaining with its reasons.
- Open the brief panel: it is a template by default and says "insufficient data" when drivers are missing.

Full limitations: `docs/LIMITATIONS.md`. Pitch outline: `docs/PITCH.md`. Code: github.com/whiskeyboo/airbus_challenge.